Dimensional Structural Inertia Law in IDT

A Consequence of Fisher Geometry and Admissibility Binding

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Abstract

In Information-Dynamic Theory (IDT), system regimes evolve under the natural gradient flow $d\omega/d\tau = -G^{-1}\nabla\Phi$ on a Fisher information manifold. The structural inertia $m^{IDT}(u) = (u^tGu)/(u^tHu)$ measures resistance to regime perturbation and equals the relaxation time of linearised dynamics (IDT-SI Theorem 1). The IDT-D3 dimensional selection program established that stable bound regimes exist uniquely in three spatial dimensions for the effective potential $\Phi_eff(r) = A/r^2 - B/r^{(d-2)}$.

This paper derives the Dimensional Structural Inertia Law: combining these two results yields the equilibrium radial inertia

$$m^*(d) = C\, r^{*2} / [2A(4 - d)]$$

where r* is the equilibrium radius and C is the Fisher stiffness constant of the scale-covariant family. Equivalently, eliminating r*:

$$m^*(d) = C/(2A(4-d)) \cdot (2A/((d-2)B))^{(2/(4-d))}$$

This law implies: only $d = 3$ admits finite positive equilibrium structural inertia; $d \to 4$ is an inertial divergence threshold; $d > 4$ has no stable bound regime. The result provides a geometric interpretation of inertia as the ratio of information-geometric stiffness to effective potential curvature.

Keywords: information geometry; structural inertia; dimensional selection; Fisher-Rao metric; gradient flow; IDT-D3; effective potential.

0. Introduction

Two lines of the IDT program have been developed separately. The structural inertia paper (IDT-SI) defines $m^{IDT}(u) = (u^tGu)/(u^tHu)$ and proves it equals the relaxation time at stable stationary regimes. The dimensional selection paper (IDT-D3) proves that for the effective potential $\Phi_eff(r) = A/r^2 - B/r^{(d-2)}$, stable bound regimes exist uniquely in $d = 3$.

Both results depend on the same underlying structure: the Fisher metric G and the Hessian H of Φ at a stationary point. This paper asks: what is the structural inertia at the equilibrium radius r* predicted by IDT-D3? The answer — the Dimensional Structural Inertia Law — connects the two programs into a single formula and reveals that $d = 3$ is the unique dimension admitting finite positive equilibrium inertia.

This is not a new postulate. It is a calculation: given the IDT-D3 potential and the scale-family Fisher metric $g_rr = C/r^2$, compute m*(d) at the equilibrium point. The result is fully determined.

1. Prerequisites

1.1 IDT-SI: Structural inertia

At a stable stationary point ω^* ($H > 0$, $G > 0$), structural inertia along direction u is:

$$m^{IDT}(u) = (u^t G u) / (u^t H u) = 1/\lambda_u$$

where λ_u is the relaxation rate of linearised dynamics along u (IDT-SI Theorem 1). For the radial direction $u = e_r$ in a 1D radial system, this reduces to:

$$m_r(r) = g_{rr}(r) / \Phi_{eff}''(r)$$

▶️ **Status: canon_derived** — follows from linearisation of $d\omega/d\tau = -G^{-1}\nabla\Phi$

1.2 IDT-D3: Effective potential and equilibrium radius

The effective potential is:

$$\Phi_{eff}(r) = A/r^2 - B/r^{(d-2)}, \quad A, B > 0$$

The stationary condition $\Phi_{eff}'(r^*) = 0$ gives:

$$(d-2)B \cdot r^{*(4-d)} = 2A$$

This has a solution $r^* \in (0,\infty)$ for all $d \neq 2$ and $d \neq 4$. The second derivative at r^* is:

$$\Phi_{eff}''(r^*) = 2A(4-d) / r^{*4}$$

This is positive (stable minimum) iff $d < 4$; in particular, $d = 3$ is the unique integer satisfying $d < 4$, $d \neq 2$ (IDT-D3 Theorem).

▶️ **Status: canon_derived** — conditional on Φ_{eff} class; proven in IDT-D3 Theorem

1.3 Scale-family Fisher metric

For regular one-parameter scale families $p(x|r) = (1/r)f(x/r)$, the Fisher metric satisfies:

$$g_{rr}(r) = C_f / r^2, \quad C_f > 0$$

This is theorem-level for regular scale families (IDT-D3 Appendix F). The identification of g_{rr} with the confinement geometry is a strengthened consistent extension.

▶️ **Status: strengthened_consistent_extension** — $g_{rr} = C/r^2$ is theorem-level for scale families; energy identification remains a closure

2. Theorem: Dimensional Structural Inertia Law

Theorem (Dimensional Structural Inertia Law)

Let $\Phi_{eff}(r) = A/r^2 - B/r^{(d-2)}$ with $A, B > 0$, and let $g_{rr}(r) = C/r^2$ with $C > 0$.
Let r^* satisfy $(d-2)B \cdot r^{*(4-d)} = 2A$ (equilibrium condition).
Then the structural inertia at equilibrium along the radial direction is:

$$m^*(d) = C r^{*2} / [2A(4 - d)]$$

Equivalently, eliminating r^*:

$$m^*(d) = C/(2A(4-d)) \cdot (2A / ((d-2)B))^{(2/(4-d))}$$

Consequences:

(i) $d = 3$: $m^*(3) = 2CA/B^2 > 0$ (finite positive)
(ii) $d \to 4$: $m^*(d) \to +\infty$ (inertial divergence)
(iii) $d > 4$: no stable bound minimum; $m^*(d) < 0$ (unphysical)
(iv) $d \leq 2$: no bound state ($d=2$ logarithmic; $d \leq 1$ no admissibility binding)

3. Proof

Step 1: Compute Φ_eff″(r*)

From Φ_eff(r) = A/r² − B/r^(d−2), differentiate twice:

```
Φ_eff′(r)   = −2A/r³ + (d−2)B/r^(d−1)
Φ_eff″(r)   =   6A/r⁴ − (d−2)(d−1)B/r^d
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At the stationary point, the equilibrium condition (d−2)B = 2A·r*^(d−4) gives:

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(d−2)(d−1)B / r*^d  =  (d−1) · (d−2)B / r*^d
                    =  (d−1) · 2A · r*^(d−4) / r*^d  =
 2A(d−1)/r*⁴
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Substituting:

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Φ_eff″(r*)  =  6A/r*⁴ − 2A(d−1)/r*⁴  =  2A[3−(d−1)]/r*⁴  =
 2A(4−d)/r*⁴
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Step 2: Compute m*(d)

Substituting g_rr(r*) = C/r*² and Φ_eff″(r*) = 2A(4−d)/r*⁴:

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m*(d)  =  g_rr(r*) / Φ_eff″(r*)  =  [C/r*²] / [2A(4−d)/r*⁴]
       =  C · r*⁴ / [r*² · 2A(4−d)]  =  C r*² / [2A(4−d)]   ∎
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Step 3: Explicit formula (r* eliminated)

From the stationary condition: r*² = (2A/((d−2)B))^(2/(4−d)). Substituting:

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m*(d)  =  C/(2A(4−d)) · (2A/((d−2)B))^(2/(4−d))
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For d = 3: exponent 2/(4−3) = 2, so r*² = (2A/B)², and:

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m*(3)  =  C/(2A·1) · (2A/B)²  =  C · 4A² / (2A B²)  =  2CA/B²
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Step 4: Sign analysis

The sign of m*(d) is determined by the sign of (4−d):

```
4−d > 0  ⇔  d < 4   ⇒  m*(d) > 0   (physically meaningful)
4−d = 0  ⇔  d = 4   ⇒  m*(d) = +∞   (divergence)
4−d < 0  ⇔  d > 4   ⇒  m*(d) < 0   (no stable bound regime)
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The last case is consistent with IDT-D3: for d > 4 the stationary point is a maximum, not a minimum, so structural inertia is ill-defined.

4. Illustration

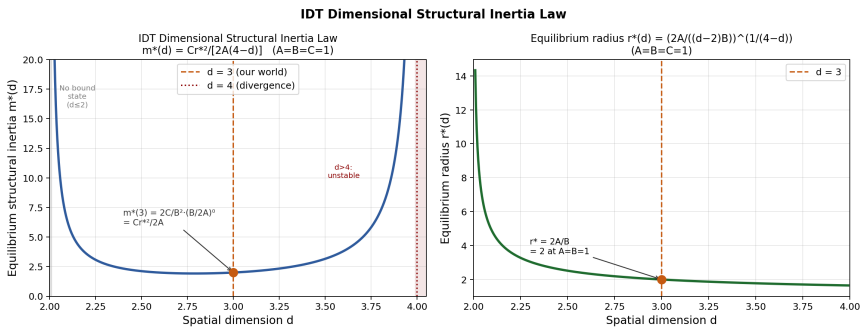

Figure 1. Left: equilibrium structural inertia m(d) for d ∈ (2, 4), A = B = C = 1. The inertia is finite and positive throughout (2, 4), diverges as d → 4, and d = 3 is the unique integer in the stable range. Right: equilibrium radius r*(d).*

5. Numerical Verification

At A = B = C = 1:

d	r*	m*(d)	Status
1	—	—	No bound state (d−2 = −1)
2	—	—	Logarithmic binding, no finite r*
3	2.000	**2.000**	**Finite positive ✓**
4	—	**+∞**	Inertial divergence
5	1.500	**−1.125**	m* < 0, no stable minimum

6. Interpretation

6.1 Inertia as curvature ratio

The formula $m^*(d) = g_{rr}(r^*)/\Phi_{eff}''(r^*)$ has a clear geometric meaning:

Numerator $g_{rr}(r^*) = C/r^{*2}$: the Fisher stiffness of the information geometry — how much the distribution changes per unit displacement in parameter space.

Denominator $\Phi_{eff}''(r^*)$: the curvature of the effective potential — how strongly the regime is confined near equilibrium.

Structural inertia is high when the regime is geometrically 'flat' (large Fisher stiffness relative to potential curvature). It is the informational analogue of inertial mass: resistance to change normalised by sensitivity.

6.2 Why only d = 3

The uniqueness of d = 3 is not imposed — it emerges from two independent conditions:

(A) A stable minimum requires $\Phi_{eff}''(r^*) > 0$, which requires $d < 4$ (IDT-D3).

(B) An integer dimension avoiding the degenerate cases d = 2 (logarithmic binding) and d = 4 (marginal).

Condition (A) alone allows any real $d \in (2, 4)$. Adding (B) singles out d = 3 as the unique integer solution. The Dimensional Structural Inertia Law quantifies this: m*(d) is finite and positive throughout (2, 4), but d = 3 is the only integer giving a physically complete bound regime.

6.3 Relation to physical mass (speculative)

The formula $m^*(3) = 2CA/B^2$ has dimensions of [length²] in natural units (C dimensionless, A and B from the potential). If r* is identified with a characteristic length scale of the bound regime, then m* scales as that length squared — a relation reminiscent of the de Broglie wavelength or Compton radius in quantum mechanics.

This connection is speculative. It requires: (i) closing the $\tau \to t$ gap (IDT GAP-3); (ii) canonical derivation of A and B from IDT axioms. Neither is currently available.

▶️ **Status: conjecture** — mass connection; requires $\tau \to t$ bridge (IDT GAP-3)

7. Status of Claims

Claim	Status	Note
m_IDT = relaxation time (IDT-SI Thm 1)	**canon_derived**	*Proved from linearised gradient flow*
$\Phi_{eff}''(r^*) = 2A(4-d)/r^{*4}$	**canon_derived**	*Pure calculus from IDT-D3 potential*

m*(d) = Cr*²/[2A(4-d)]	**canon_derived**	*Given P1a + P1b (see below)*
g_rr = C/r² (scale families)	**strengthened_ext.**	*Theorem-level for regular scale families*
Confinement term = Fisher stiffness	**strengthened_ext.**	*Energy identification; closure step*
Admissibility transport → 1/r^(d-2)	**strengthened_ext.**	*Fick's law closure; not micro-derived*
m*(3) ~ physical mass	**conjecture**	*Requires τ→t bridge (GAP-3)*

8. Open Problems

GAP-E1: Canonical derivation of g_rr = C/r² from IDT axioms A0–A5 (not just scale-family theorem).

GAP-E2: Microscopic derivation of admissibility transport J = −D∇$\mathcal{M}$ from regime dynamics.

GAP-E3: Canonical derivation of A and B from IDT axioms, giving absolute values of m*(3).

GAP-3 (IDT program): Identify IDT internal time τ with physical time t. Critical for connecting m*(3) to physical mass.

If GAP-E1 and GAP-E2 are closed, the Dimensional Structural Inertia Law upgrades from 'conditional canon-derived' to 'fully canon-derived'. If additionally GAP-3 is closed, the connection to physical mass becomes a theorem-level statement.

9. Conclusion

We have derived the Dimensional Structural Inertia Law:

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m*(d)  =  C r*² / [2A(4−d)]
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by combining the IDT-SI relaxation-time interpretation of structural inertia with the IDT-D3 effective potential and the scale-family Fisher geometry. The law is a direct calculation with no new postulates beyond those already present in the two parent papers.

The result shows that d = 3 is the unique integer dimension admitting finite positive equilibrium structural inertia. This connects the geometric notion of inertia-as-resistance-to-perturbation to the information-geometric selection of three spatial dimensions.

The derivational status is honest: the theorem is canon-derived conditional on the closure assumptions (P1a, P1b) inherited from IDT-D3. Closing these gaps remains the primary open problem of the IDT program.

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Appendix: Key Formulas Summary

Effective potential:

$$\Phi_{eff}(r) = A/r^2 - B/r^{(d-2)}$$

Equilibrium condition:

$$(d-2)B \cdot r^{*(4-d)} = 2A \quad \Rightarrow \quad r^* = (2A/((d-2)B))^{(1/(4-d))}$$

Second derivative at equilibrium:

$$\Phi''_{eff}(r^*) = 2A(4-d)/r^{*4}$$

Fisher metric (scale families):

$$g_{rr}(r) = C/r^2$$

Structural inertia at equilibrium:

$$m^*(d) = g_{rr}(r^*) / \Phi''_{eff}(r^*) = C\, r^{*2} / [2A(4-d)]$$

Fully explicit (r* eliminated):

$$m^*(d) = C/(2A(4-d)) \cdot (2A/((d-2)B))^{(2/(4-d))}$$

Special case d = 3:

$$m^*(3) = 2CA/B^2$$